Huawei ICT Competition 2025–2026
Innovation Competition

Entry Information Form

Team Information			
University Name	Misr University for Science & Technology (MUST)		
Team Name	Learnify		
Team Members	**Instructor**	**Students**	
	Eng.Ahmed Tarek	1.Shahd Emad	2.Raghd Tamer | 3.Malak Tamer
Team Captain Tel	01010701735		
Team Captain Email	shahdmarouf668@gmail.com		
Team Group Photo	*Please provide one team photo of about 2 MB in size.*		

Entry Information	
Topic	Developing AI innovation applications powered by MindSpore – Education Sector
Entry Name	Learnify: Intelligent Learning Platform for University Students
Applicable Fields	Education, Artificial Intelligence, and Information Technology
Technologies Used	MindSpore, ModelArts (primary); OpenAI API, TensorFlow, PyTorch, Scikit-learn, Python, JS, HTML, CSS (supporting tools)
Entry Introduction	**1. Background and Value** In today's educational environment, university students face challenges in managing their course materials, tracking academic progress, and preparing effectively for exams. Learnify aims to provide an intelligent platform that simplifies learning through integrated tools like material management, automated quiz generation, GPA calculation, note-taking, and chatbot assistance. This project is relevant to the IT field because it leverages artificial intelligence, web technologies, and user-centered design to improve the learning experience. The platform addresses the need for a smart system that integrates academic tools in one place, which is a struggle for students. to the digital transformation of education, improving productivity and learning outcomes across diverse academic disciplines.

2. Introduction - Innovative Aspects

Learnify's innovative aspect lies in integrating multiple academic support tools into a single, intelligent, web-based platform. The platform uses **AI for generating customized quizzes** and providing instant student support via an **AI chatbot**. It also features a **ranking system based on quiz performance** to introduce a competitive element for motivation. The comprehensive feature set includes organized material access, GPA calculation, and digital note-taking

Technical Solution Introduction

1. Technical Solution Introduction

Learnify is a web-based platform where students can view materials, generate quizzes, calculate GPA, take notes, and chat with an AI assistant. The overall system tracks performance and ranks users based on quiz results. **Technology Stack:** Frontend: Vue.js, Backend: .NET, Database: MySQL, **AI: OpenAI API, TensorFlow, PyTorch**. The development follows the **Agile methodology**.

2. Key Technical Modules

Module 1: AI-Powered Quiz Generator: Uses AI (OpenAI API, TensorFlow, PyTorch) to generate customized quizzes

Module 2: AI Chatbot: An intelligent chatbot for instant help and student support.

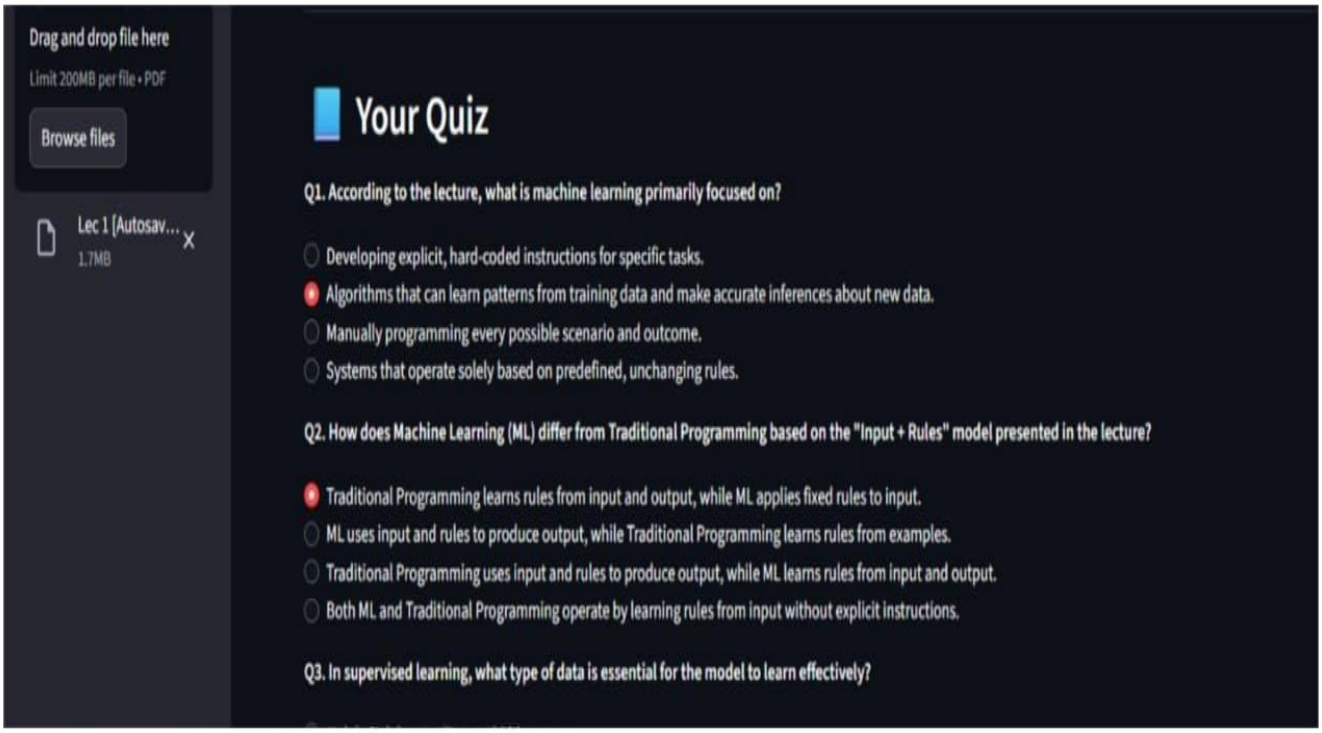
Module 3: Data Management & Performance Tracking: Handles data from user-generated content and educational resources (Data Sources). Manages course materials , calculates GPA , and tracks performance for the ranking system.

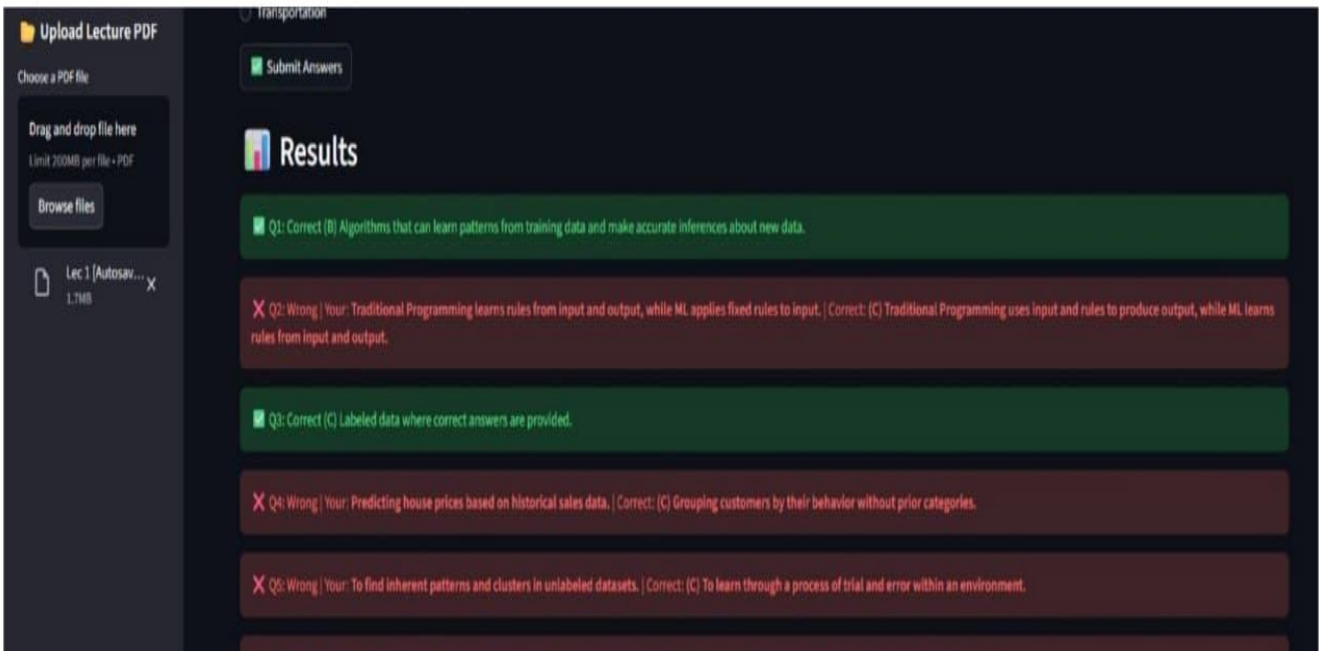
3. Final Achievements/Metrics

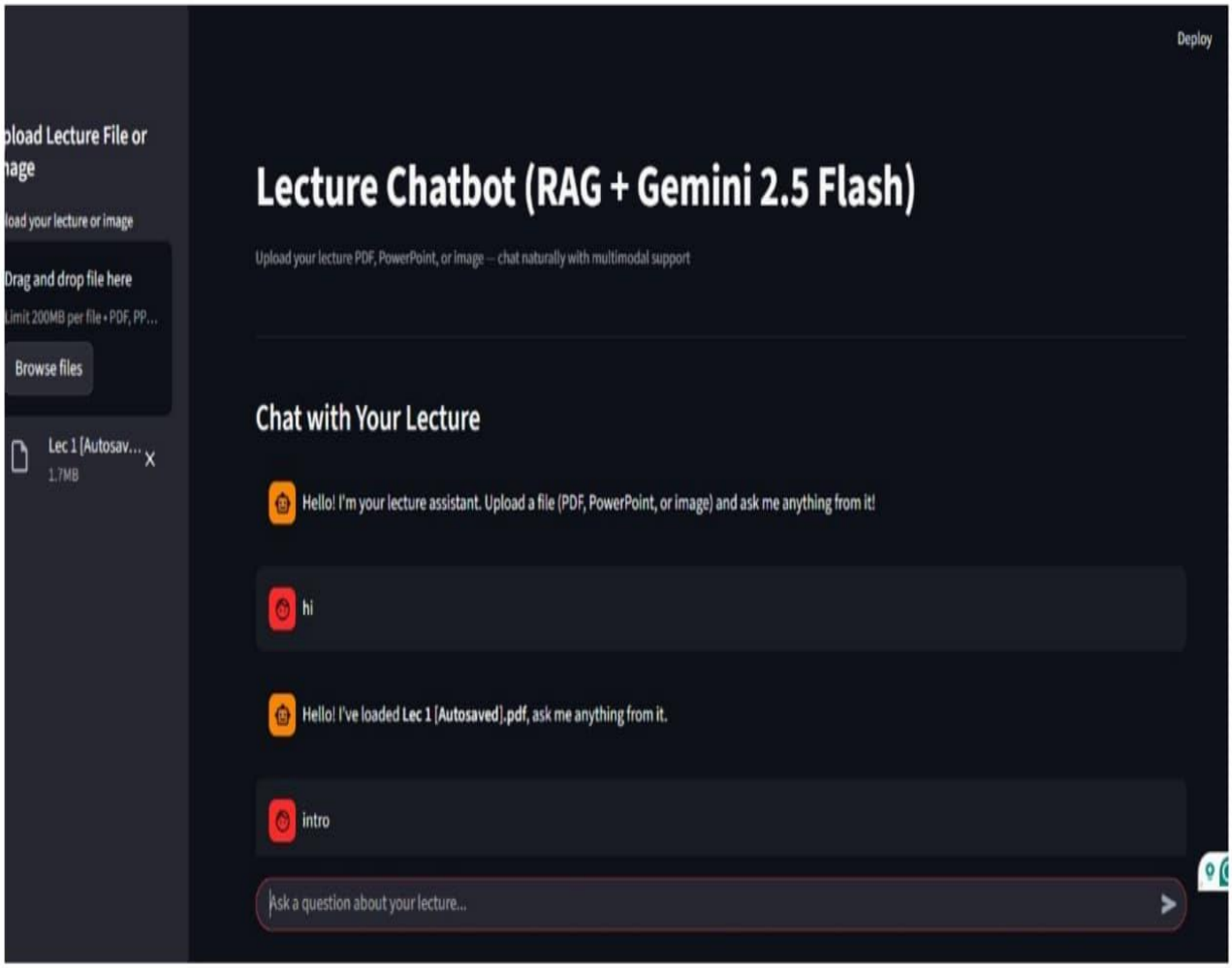
A fully functional website with features including: AI-powered quiz generation and automatic grading , GPA calculation and progress tracking , and a performance

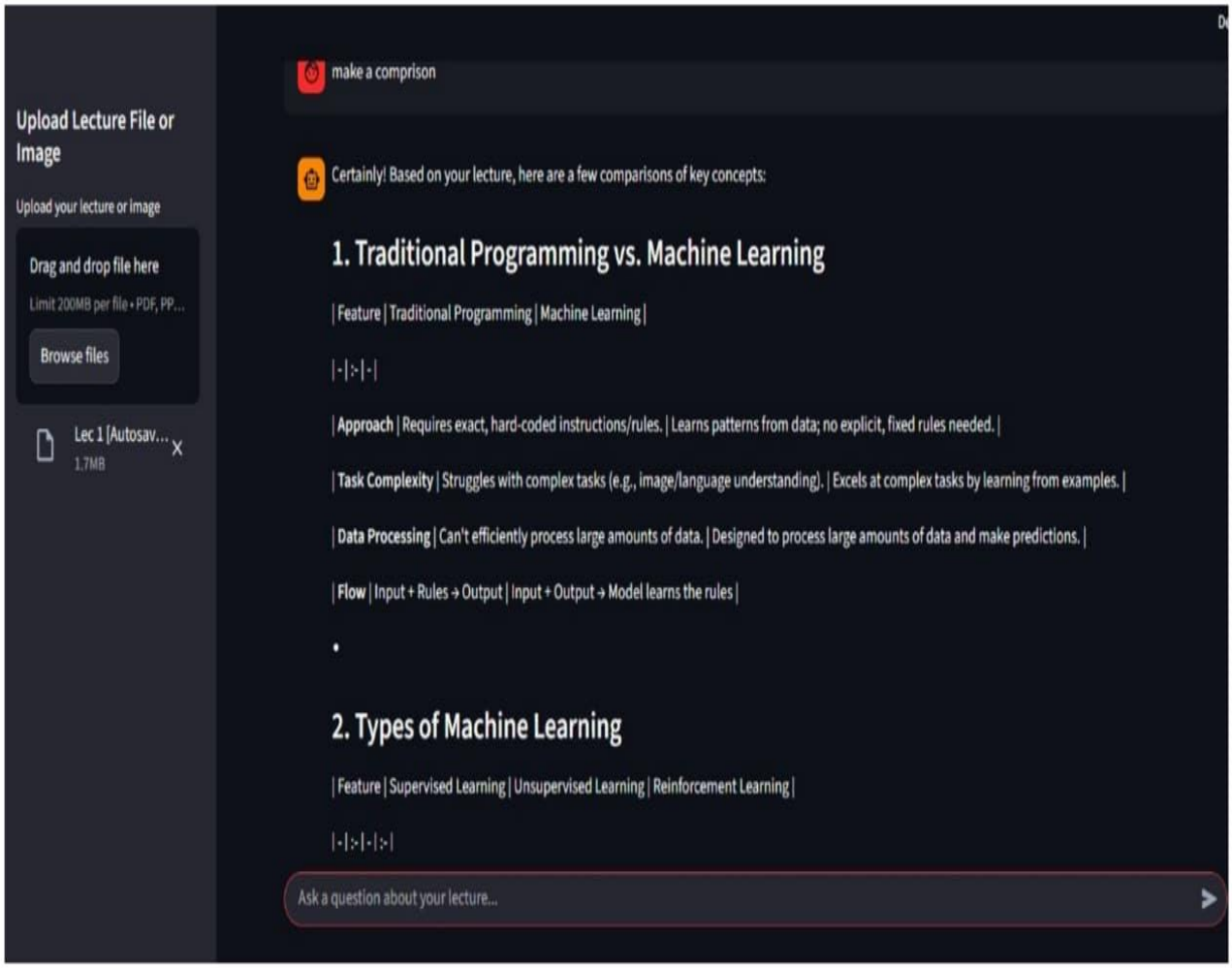
	ranking/leaderboard system. **Measurable Metrics (Examples): Accuracy** (e.g., Quiz generation relevance, GPA calculation accuracy), **Inference Throughput** (e.g., Chatbot response time, Quiz generation speed), **Expected Cost Reduction** (e.g., Reduced need for manual teaching assistant support for quizzes/basic inquiries)
Entry Demonstration	

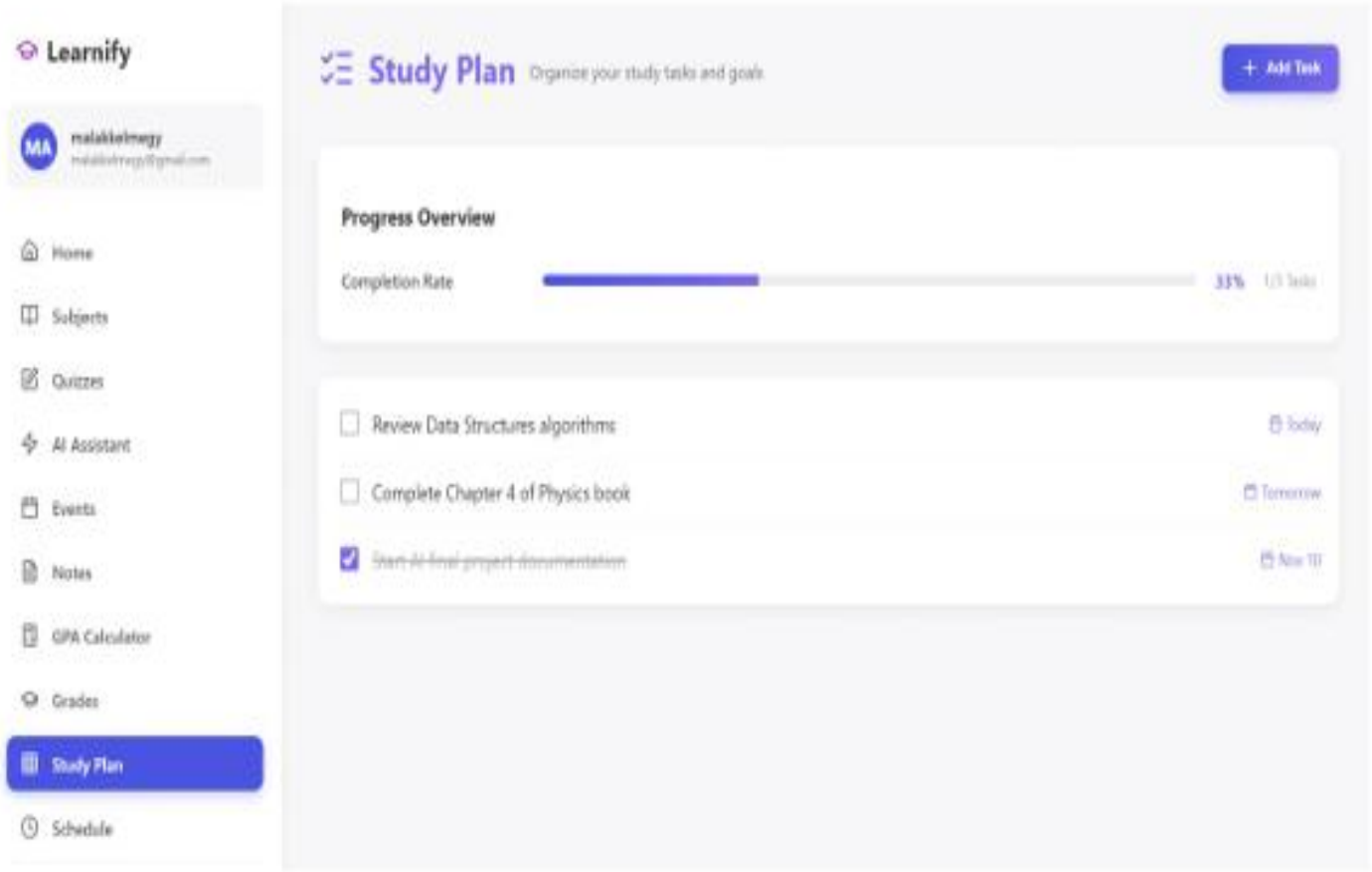

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Generate Quiz
Enter a topic (optional):
e.g., Syntax, Semantics, Grammar...
Generate Quiz
Generating your quiz...

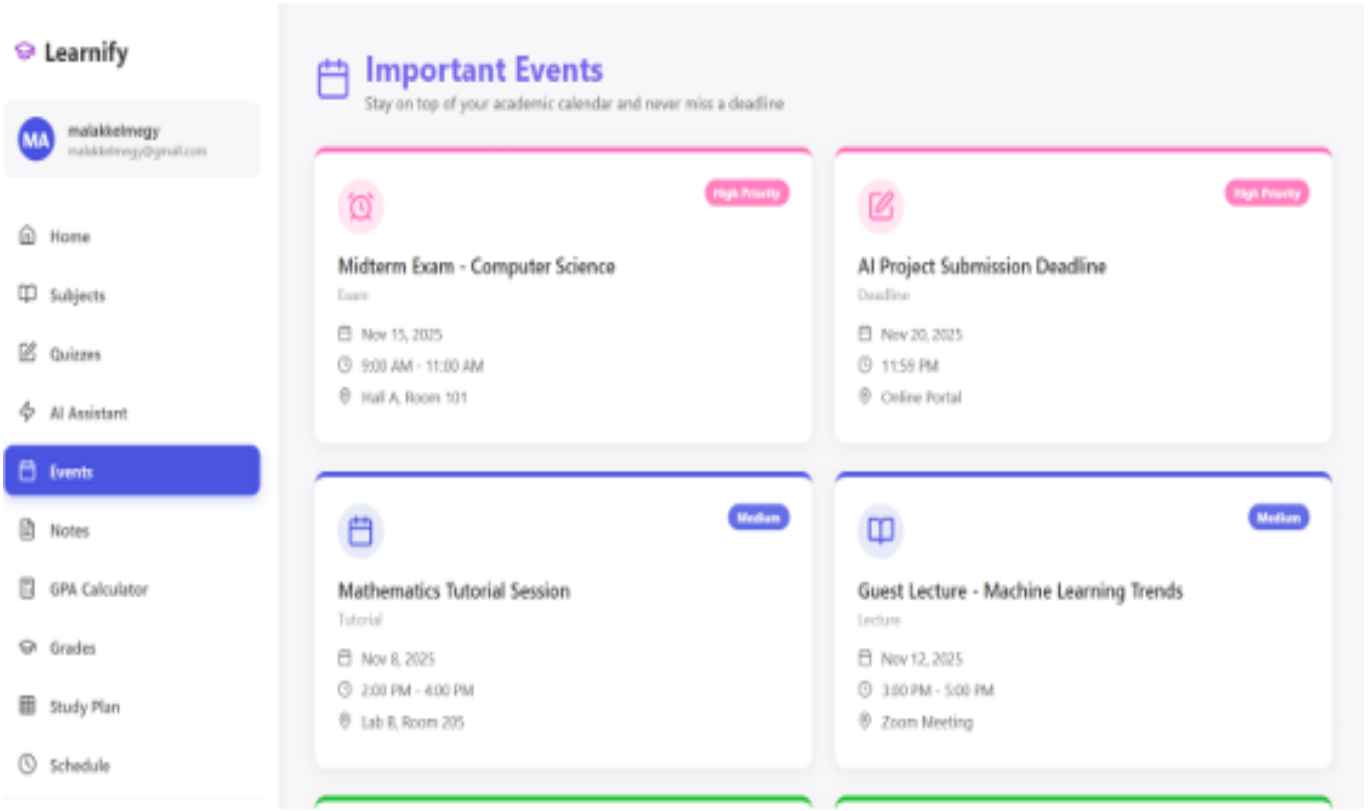

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Your Quiz
Q1. According to the lecture, what is machine learning primarily focused on?
Developing explicit, hard-coded instructions for specific tasks.
Algorithms that can learn patterns from training data and make accurate inferences about new data.
Manually programming every possible scenario and outcome.
Systems that operate solely based on predefined, unchanging rules.
Q2. How does Machine Learning (ML) differ from Traditional Programming based on the "Input + Rules" model presented in the lecture?
Traditional Programming learns rules from input and output, while ML applies fixed rules to input.
ML uses input and rules to produce output, while Traditional Programming learns rules from examples.
Traditional Programming uses input and rules to produce output, while ML learns rules from input and output.
Both ML and Traditional Programming operate by learning rules from input without explicit instructions.
Q3. In supervised learning, what type of data is essential for the model to learn effectively?

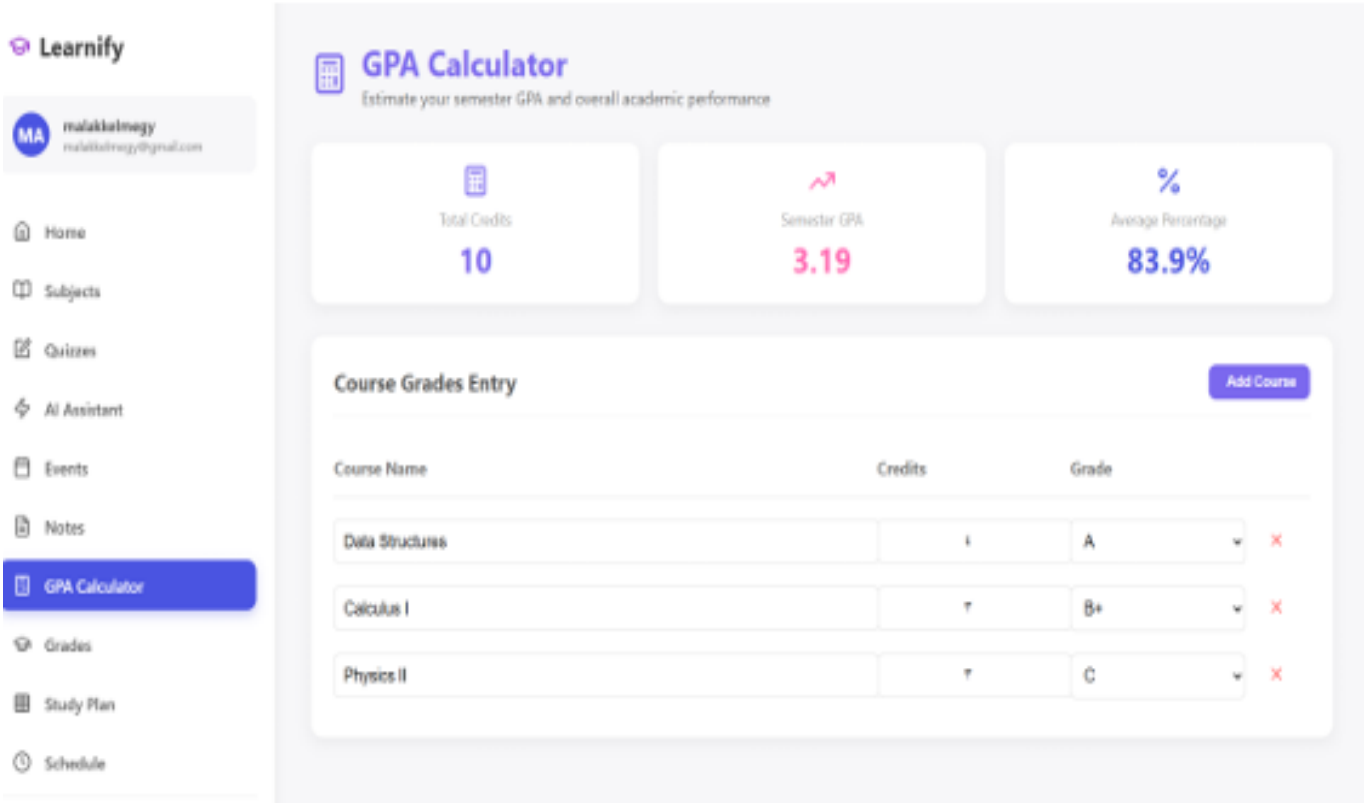

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Results
Q1: Correct (B) Algorithms that can learn patterns from training data and make accurate inferences about new data.
Q2: Wrong | Your: Traditional Programming learns rules from input and output, while ML applies fixed rules to input. | Correct: (C) Traditional Programming uses input and rules to produce output, while ML learns rules from input and output.
Q3: Correct (C) Labeled data where correct answers are provided.
Q4: Wrong | Your: Predicting house prices based on historical sales data. | Correct: (C) Grouping customers by their behavior without prior categories.
Q5: Wrong | Your: To find inherent patterns and clusters in unlabeled datasets. | Correct: (C) To learn through a process of trial and error within an environment.

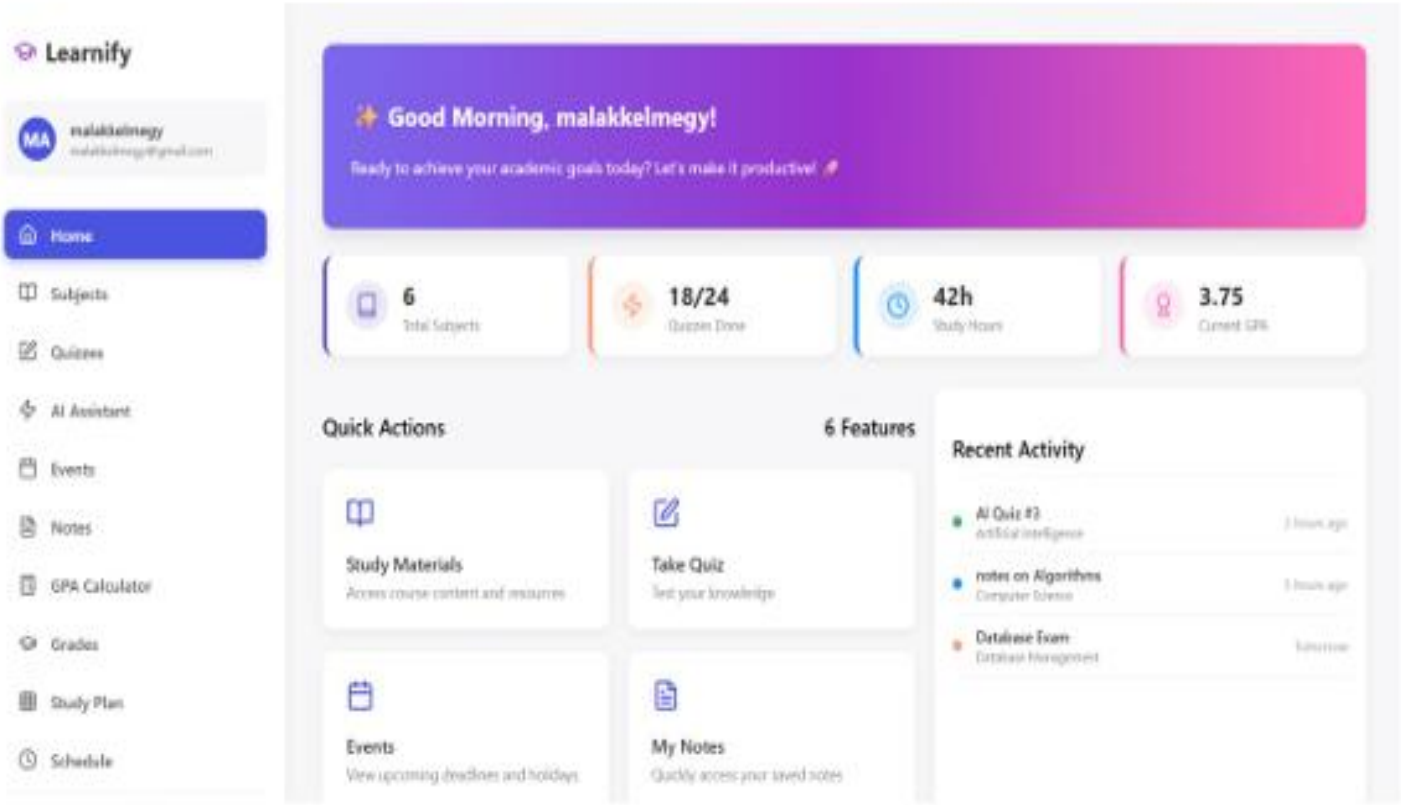

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intro
Ask a question about your lecture...

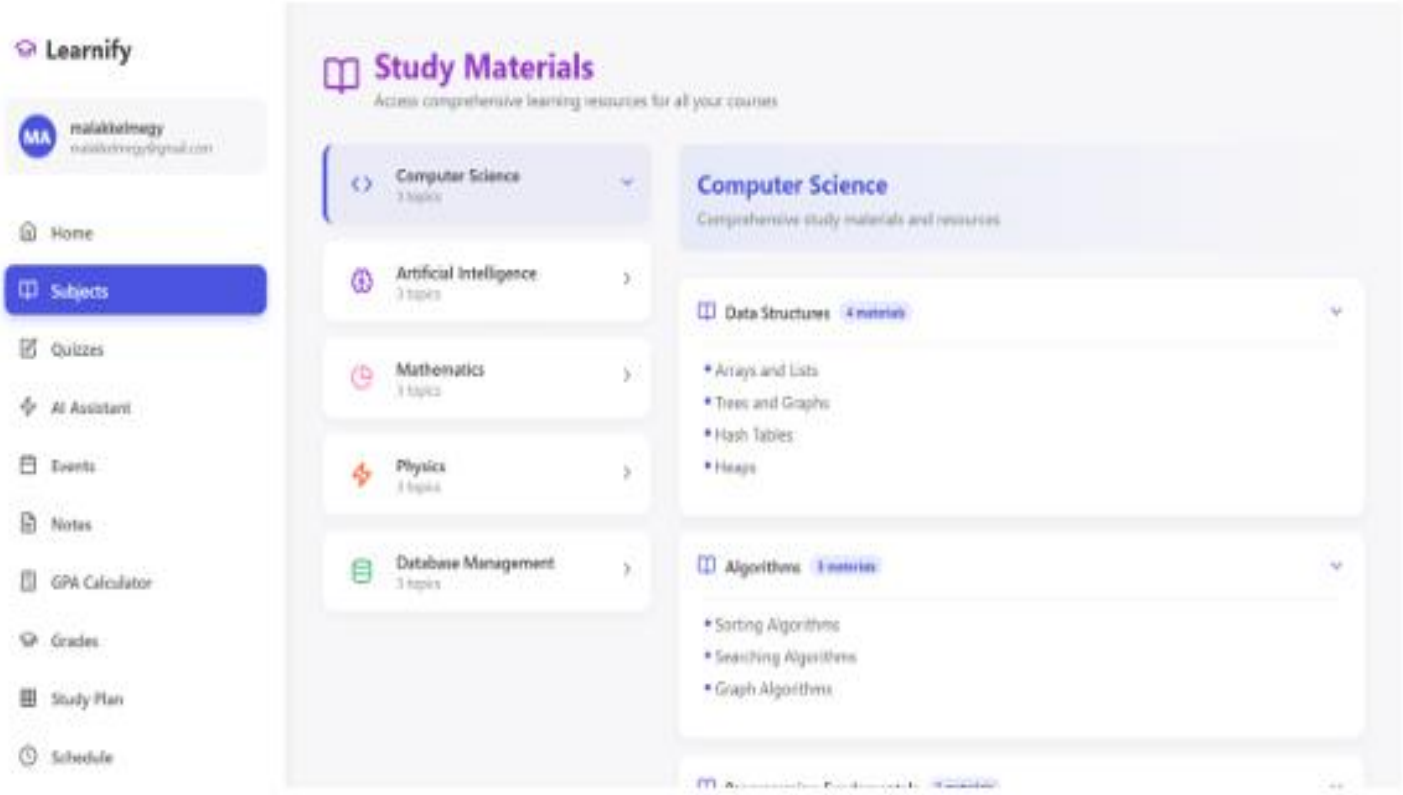

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Certainly! Based on your lecture, here are a few comparisons of key concepts:
1. Traditional Programming vs. Machine Learning
Feature	Traditional Programming	Machine Learning
Approach	Requires exact, hard-coded instructions/rules.	Learns patterns from data; no explicit, fixed rules needed.
Task Complexity	Struggles with complex tasks (e.g., image/language understanding).	Excels at complex tasks by learning from examples.
Data Processing	Can't efficiently process large amounts of data.	Designed to process large amounts of data and make predictions.
Flow	Input + Rules → Output	Input + Output → Model learns the rules
2. Types of Machine Learning
Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Ask a question about your lecture...


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Study Plan
Schedule
Study Plan
Organize your study tasks and goals
+ Add Task
Progress Overview
Completion Rate
33%
Review Data Structures algorithms
Today
Complete Chapter 4 of Physics book
Tomorrow


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Important Events
Stay on top of your academic calendar and never miss a deadline
High Priority
Midterm Exam - Computer Science
Exam
Nov 15, 2025
9:00 AM - 11:00 AM
Hall A, Room 101
High Priority
AI Project Submission Deadline
Deadline
Nov 20, 2025
11:59 PM
Online Portal
Medium
Mathematics Tutorial Session
Tutorial
Nov 8, 2025
2:00 PM - 4:00 PM
Lab B, Room 205
Medium
Guest Lecture - Machine Learning Trends
Lecture
Nov 12, 2025
3:00 PM - 5:00 PM
Zoom Meeting


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Estimate your semester GPA and overall academic performance
Total Credits
10
Semester GPA
3.19
Average Percentage
83.9%
Course Grades Entry
Add Course
Course Name
Credits
Grade
Data Structures
A
Calculus I
B+
Physics II
C


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Recent Activity
AI Quiz #3
notes on Algorithms
Database Exam


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Access comprehensive learning resources for all your courses
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Artificial Intelligence
Mathematics
Physics
Database Management
Computer Science
Comprehensive study materials and resources
Data Structures
Arrays and Lists
Trees and Graphs
Hash Tables
Heaps
Algorithms
Sorting Algorithms
Searching Algorithms
Graph Algorithms